Machine Learning-Based Motion Intent Recognition for Assistive Robotics

Michal Ridner[1] Onur Ozkaya[2] Romina Bahrami[3]

[1]Model Exploration and Implementation + Feature Ablation

[2]EDA + Model Exploration and Implementation

[3]Model Exploration, Optimization, and Explainability

[1,3]Department of Electrical and Computer Engineering

[2]Department of Mechanical and Industrial Engineering

Introduction to Machine Learning – Winter 2026

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Project Overview

Our goal is to implement and evaluate machine learning models that recognize human motion intent (e.g., walking, sitting, standing) from wearable sensor data to support smarter assistive robotic systems such as exoskeletons.

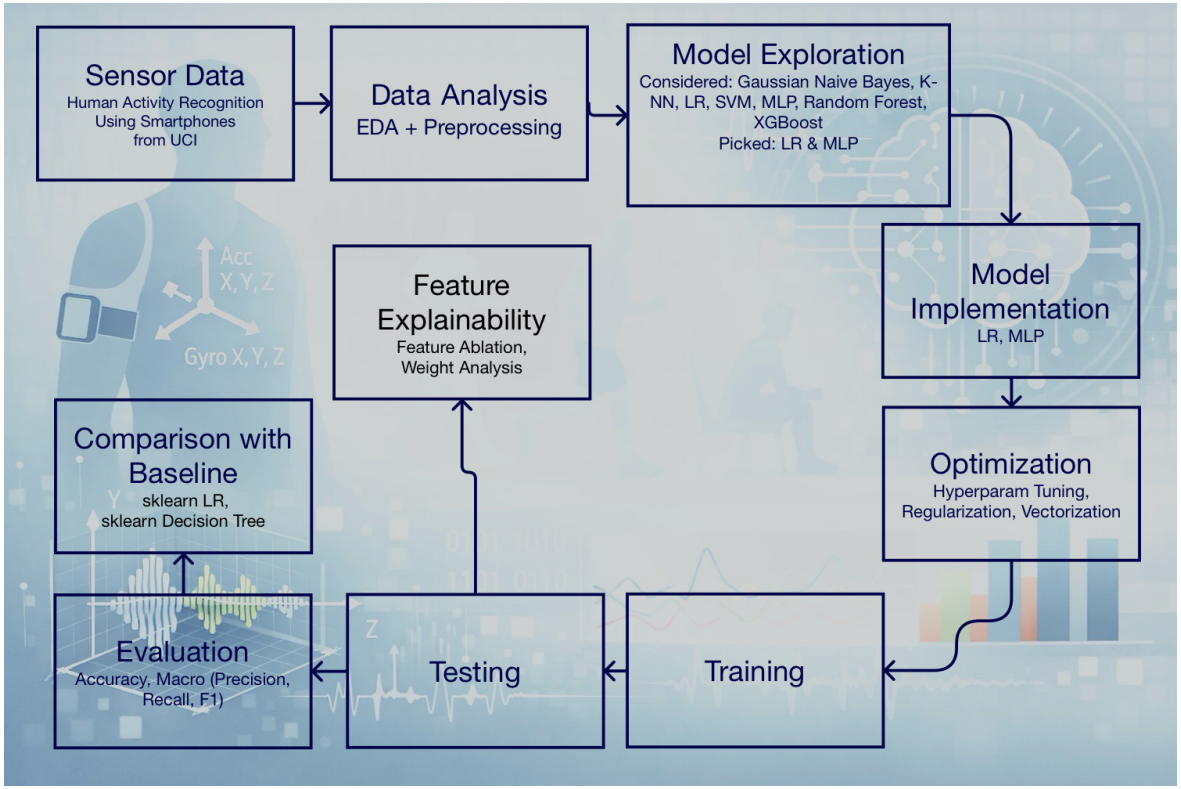

Figure: Demonstration of our workflow

Data Analysis

6 activities:

- Walking
- Walking_upstairs
- Walking_downstairs
- Sitting
- Standing
- Laying

Figure: Axes orientation [1]

561 standardized features, with 4 main feature subsets

- Sensor: Gyroscope/Accelerometer
- Component: Body/Gravity
- Domain: Time/Frequency
- Axis: X/Y/Z/Multi-Axis

[1] Anguita, Davide & Ghio, Alessandro & Oneto, Luca & Parra, Francesc & Reyes-Ortiz, J.L.. (2013). Energy efficient smartphone-based activity recognition using fixed-point arithmetic. Journal of Universal Computer Science. 19. 1295-1314.
[2] Data source: Human Activity Recognition Using Smartphones Dataset from UC Irvine ML Repository

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Model Development

Logistic Regression

- Cross-entropy loss function
- Softmax with one-hot encoding
- Full-batch gradient descent
- Optimization
  - Hyperparameter tuning (max_iter, learning_rate)
  - Regularization (penalizing large max_iter, small learning_rate)
  - Vectorization for faster computation
- Achieves 94.3% test accuracy

MLP

- Cross-entropy loss function
- Label encoding
- Adam optimizer
- Optimization
  - Hyperparameter tuning (model type, batch_size, learning_rate)
  - Regularization (Dropout, weight_decay)
- Achieves 95.45% test accuracy

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Summary of Findings

Test-set Results (Accuracy / Macro F1)

Model	Full Features	After PCA
Decision Tree (rule-like partition)	0.863 / 0.860	0.756 / 0.753
sklearn Logistic Regression	0.951 / 0.951	0.925 / 0.924
Our Logistic Regression	0.943 / 0.944	0.923 / 0.922
MLP	0.955 / 0.954	0.919 / 0.919

- Strong linear models performed best overall. Our LR's performance matched the sklearn baseline closely (0.7% lower).
- The MLP matched the strongest linear baseline, showing only a small gain despite added complexity.
- Linear models remained relatively robust after PCA ($561 \rightarrow 67$), but the Decision Tree dropped sharply, indicating PCA preserves linear structure better than tree-friendly feature structure.
- Feature analysis supports interpretability. Time-domain features alone nearly matched the full feature set. Sitting vs. standing is hardest to predict.

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